

Hippocampal processed traces: a bounded population-state forecast

The source archive contains 157 MAT sessions from 28 animal suffixes across dorsal/ventral CA1, contexts and study days. Twenty-seven Day 4 files have no processed.trace endpoint, leaving 130 trace sessions. All sessions from each animal are linked: 23 development animals and five reserved animals. A schema-inspected animal was permanently assigned development before numerical fitting.

Endpoint and selected equation

The task predicts the source-component population mean over native frames $t + 31$ through $t + 60$. Let e_t be the current 30-frame population mean, m_t the preceding 300-frame mean, and $v_t = 2(e_t - e_{t-30})$. The selected compact equation is

$$\hat{e} = 0.31327391e_t + 0.68672609m_t - 0.088994875\max(v_t, 0) - 0.0065038607\min(v_t, 0).$$

All amplitudes remain **author processed-trace units** and all times remain native frame indices. The dimensionless coefficients describe released-signal recovery/asymmetry; they are not calcium lifetimes, spike rates or neuronal coupling constants. The current and historical summaries use only frames at or before t .

Six adaptive attempts tested uniform decay, local set-point recovery, inertia, cross-component heterogeneity, regional response and asymmetric rising/declining states. Development selected the compact asymmetric recovery relation within the 1% simplicity tolerance. All coefficient states and alternative results remain available.

Frozen results

Model	Development group error	Reserved group error
baseline_persistence	0.0002879488	0.0003369348
baseline_velocity	0.0007027432	0.00083927
baseline_flexible	0.0002585612	0.0003032324
attempt_001_decay	0.0002784271	0.0003252555
attempt_002_relaxation	0.0002456698	0.0002824797
attempt_003_inertia	0.0002449329	0.0002814723
attempt_004_heterogeneity	0.0002407902	0.0002759831
attempt_005_region	0.0002447443	0.0002818887
attempt_006_asymmetry	0.0002403833	0.0002761791

Selected before confirmation: **attempt_006_asymmetry**. Primary error is mean absolute error in author processed-trace units, averaged within each independent group and then equally across groups. All candidates are shown to preserve unfavorable results; a lower retrospective score does not change the selected reference.

The selected rule gives 0.000276179 author-trace-unit MAE on five reserved animals, compared with 0.000336935 persistence, 0.000839270 linear extrapolation and 0.000303232 flexible prediction. This is a useful scoped processed-signal forecasting extension, with no claim of a new neural law or demonstrated behavioral impact. The source does not document exact trace normalization or the polarity of exclude.SFPs flags in twelve sessions. Accordingly, all released component rows contribute; this is not an independently curated neuron-only target. Behavior arrays and validTraceFrames have unresolved indexing semantics, so freezing, speed alignment and seconds were excluded before fitting. Upstream author denoising/normalization may use future observations: causal-prefix guarantees apply to the released trace, not

real-time fluorescence acquisition. Source fear/context findings are prior art, and this forecast cannot independently confirm them.

Validation and reproducibility

Every 30 native frames after 300-frame history, forecast next 30-frame population mean ending 60 frames later. Only trace frames $\leq t$ are inputs. Native frame index is used without inferring seconds or behavior alignment. Past observed trace history is allowed. No held future calibration or source fitted maps/pvpreS/pvpostS enter features. All model parameters and flexible transforms are fitted on development animals.

Retained author-processed calcium traces only; amplitude normalization/calcium-event interpretation is not documented in the archive. No spikes, absolute calcium concentration or real-time physiology claim. Twenty-eight source animals; held animals stay within the same study. All outcomes are now exposed to later agents. Claims of new validation require fresh data or an explicitly retrospective designation. Source study and processing are disclosed in SOURCE_AND_UNITS and PRIOR_ART; this is computational confirmation, not independent experimental replication.

`python run.py` verifies the allowlisted package and reproduces all saved predictions. `rules.json` contains exact coefficients and all learned transforms; `EQUATIONS.md` names each feature and equation. The adjacent native adapter reconstructs source-hash-verified observations and predictors; `task_spec.json` declares input, timing, calibration and grouping budgets. Future proposals may use different equations under the same contract, or seek review of a genuinely different measured task.

Source: [Hippocampal processed-population trace forecasting; primary source/prior art](#). License recorded by source: CC BY4.0. Literature and semantics audited 2 October 2026.